evolution of massive stars

$M > 8 \, M_\odot$

Tuesday

Presenters	Project
Norris, Chris	Measuring Star Cluster Ages
Sarah, Josh	Estimating Stellar Properties with Asteroseismology
Jedidiah	Measuring the Rotation Rates of Stars
Gwyn	Testing Stellar Age Measurements from Isochrones
Leon, Michael	Calibrating the Initial to Final Mass Relation for White Dwarfs
Alice, Ali	Calibrating the Initial to Final Mass Relation for White Dwarfs

Thursday

Presenters	Project
Jo	Designing Surveys for the Stars
Jayanth, Sahil	The Fate of the Earth and Other Exoplanets
Will, Leo	Supernova Natal Kicks
Jacques, Matthew	The Black Hole Mass Function
Alex	The Black Hole Mass Function
Jacob, Ryan	The Black Hole Mass Function

What defines the line between massive and intermediate-mass star?

Intermediate-mass

- Does not explode at the end of life
- Leaves behind a white dwarf

Massive

- Does explode at the end of life
- Leaves behind a neutron star or black hole

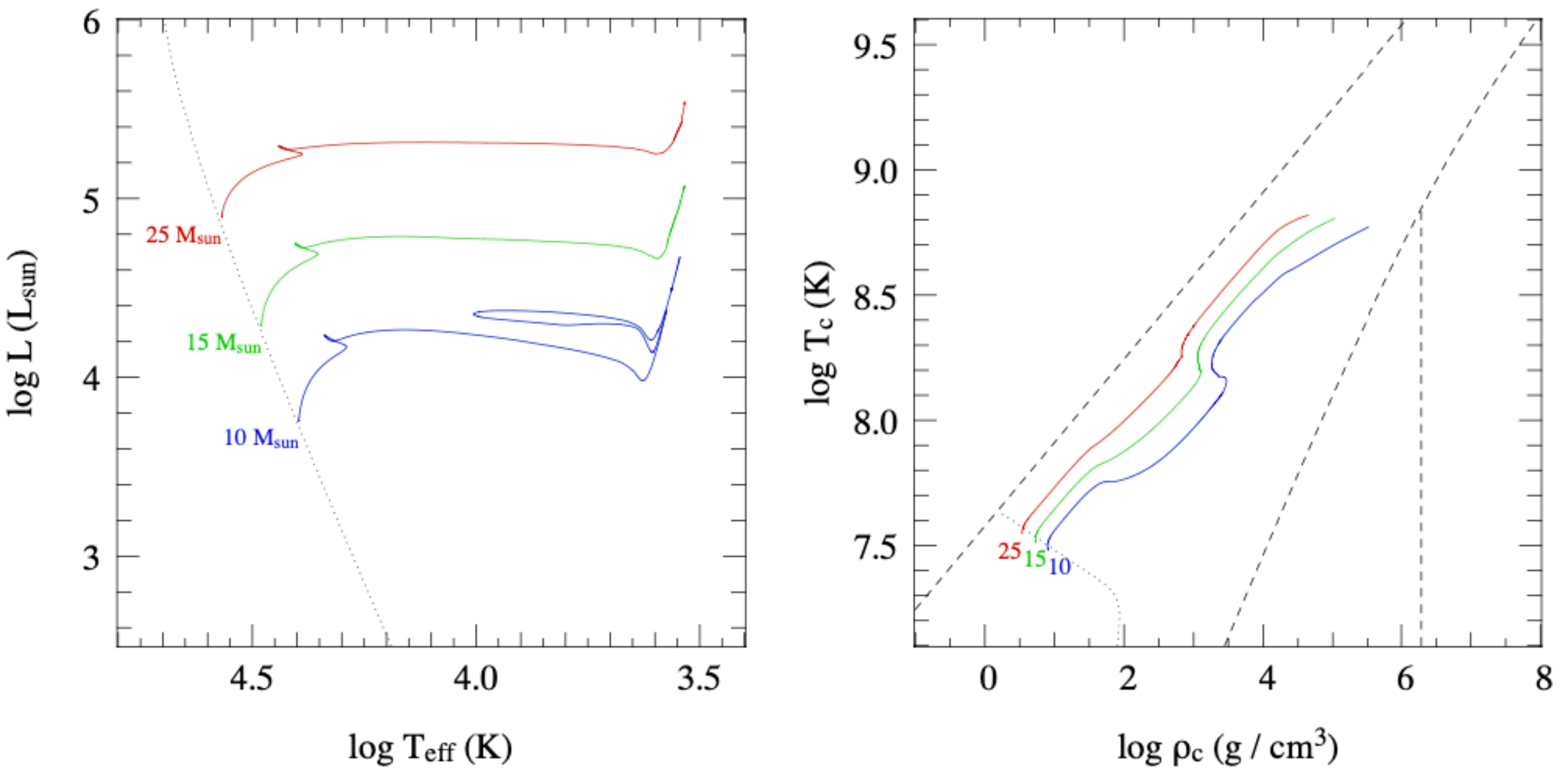
Intermediate-mass

- Does not ignite non-degenerate Carbon in the core
- Wind mass loss is only important after the main sequence is over

Massive

- Ignites non-degenerate Carbon (and potentially other elements if $M > 11M_\odot$) in the core
- Wind mass loss is important for all phases of evolution

The exact line is uncertain because we don't understand convection



$$\log(-\dot{M}) \approx -8.16 + 1.77 \log\left(\frac{L}{L_\odot}\right) - 1.68 \log\left(\frac{T_{\rm eff}}{\rm K}\right)$$

de Jager+1988: L > 1000 $L_\odot$

10 $M_\odot$ at ZAMS: $\dot{M} \simeq 10^{-9} M_\odot \mathrm{yr}^{-1}$

10 $M_\odot$ at C ignition: $\dot{M} \simeq 6 \times 10^{-7} M_\odot \mathrm{yr}^{-1}$

25 $M_\odot$ at ZAMS: $\dot{M} \simeq 5 \times 10^{-8} M_\odot \mathrm{yr}^{-1}$

25 $M_\odot$ at C ignition: $\dot{M} \simeq 2 \times 10^{-5} M_\odot \mathrm{yr}^{-1}$

Hot (OB) stars have winds driven by radiation absorption

If a photon is absorbed by an element, and all of the momentum of the photon is imparted onto that element, then the absorption carries away the stellar material with speed v_∞

The total mass loss rate is then: $\dot{M} v_\infty < \frac{L}{c}$

This is empirically observed to within a factor of 1/3 - 1/2

The largest uncertainties arise from small scale fluctuations (clumping) in stellar material that can accelerate/deccelerate the wind

Hot (OB) stars have winds driven by radiation absorption

A major consequence is that OB-star winds are *metallicity dependent*

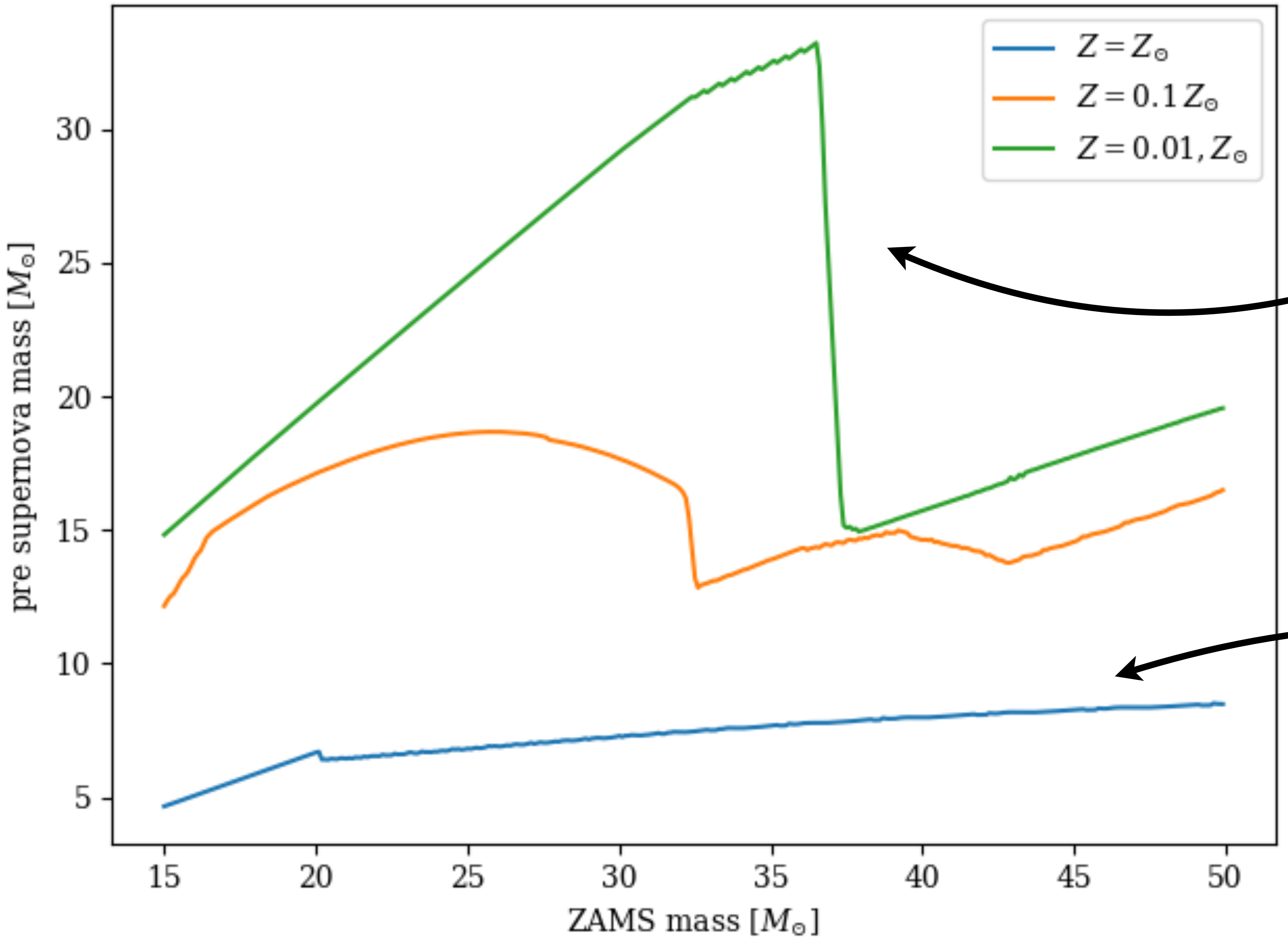
$$\begin{aligned}\log \dot{M} = & - 6.688\ (\pm 0.080) \\ & + 2.210\ (\pm 0.031)\ \log(L_*/10^5) \\ & - 1.339\ (\pm 0.068)\ \log(M_*/30) \\ & - 1.601\ (\pm 0.055)\ \log\left(\frac{v_\infty/v_{\rm esc}}{2.0}\right) \\ & + 1.07\ (\pm 0.10)\ \log(T_{\rm eff}/20\,000) \\ & + 0.85\ (\pm 0.10)\ \log(Z/Z_\odot)\end{aligned}$$

for $12\ 500 \leq T_{\rm eff} \leq 22\,500$ K

$$\begin{aligned}\log \dot{M} = & - 6.697\ (\pm 0.061) \\ & + 2.194\ (\pm 0.021)\ \log(L_*/10^5) \\ & - 1.313\ (\pm 0.046)\ \log(M_*/30) \\ & - 1.226\ (\pm 0.037)\ \log\left(\frac{v_\infty/v_{\rm esc}}{2.0}\right) \\ & + 0.933\ (\pm 0.064)\ \log(T_{\rm eff}/40\,000) \\ & - 10.92\ (\pm 0.90)\ \{\log(T_{\rm eff}/40\,000)\}^2 \\ & + 0.85\ (\pm 0.10)\ \log(Z/Z_\odot)\end{aligned}$$

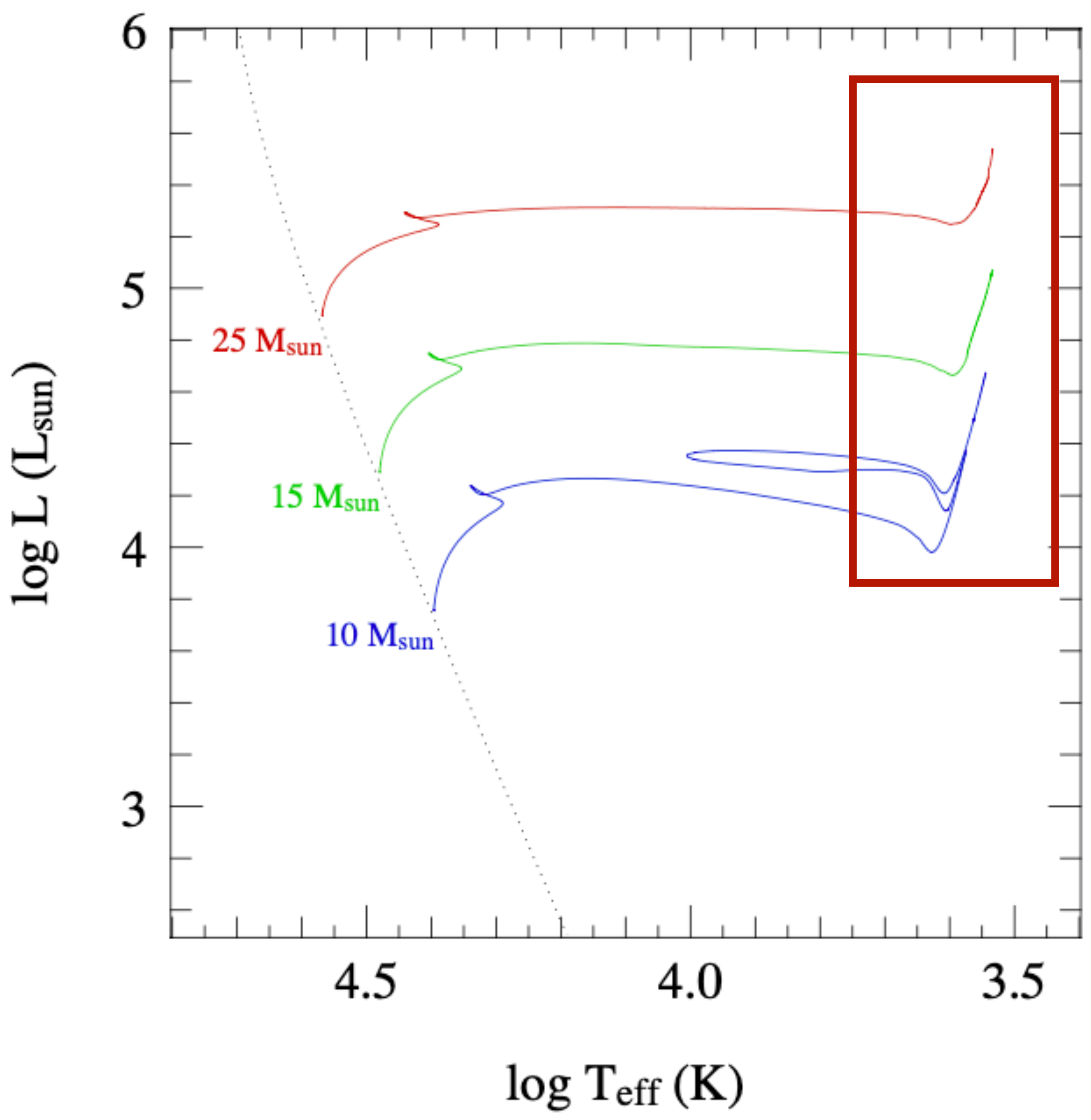
for $27\ 500 < T_{\rm eff} \leq 50\,000$ K

Vink+2001



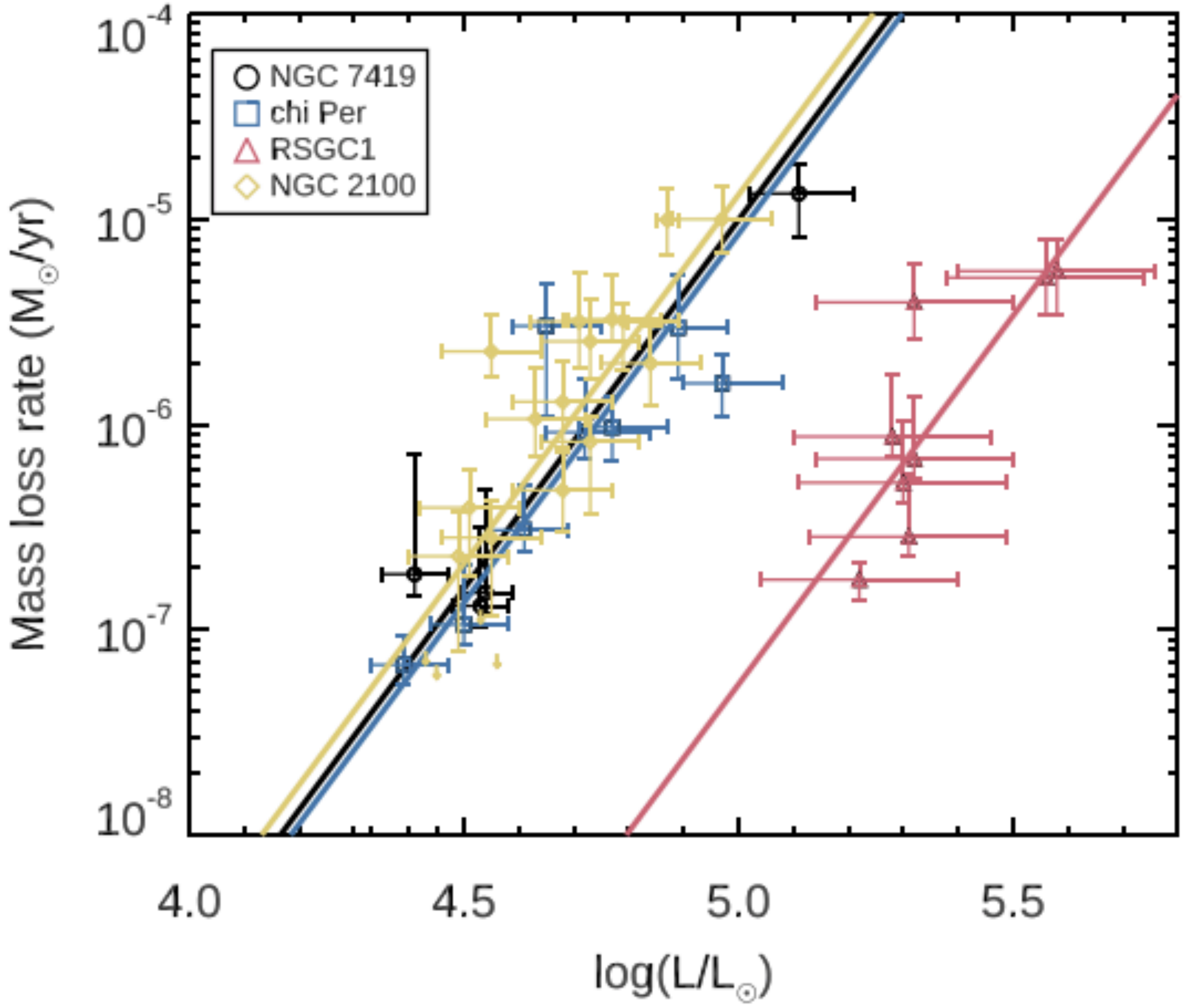
At lower metallicities, more massive stellar remnants are produced

Winds can strip almost all of the envelope

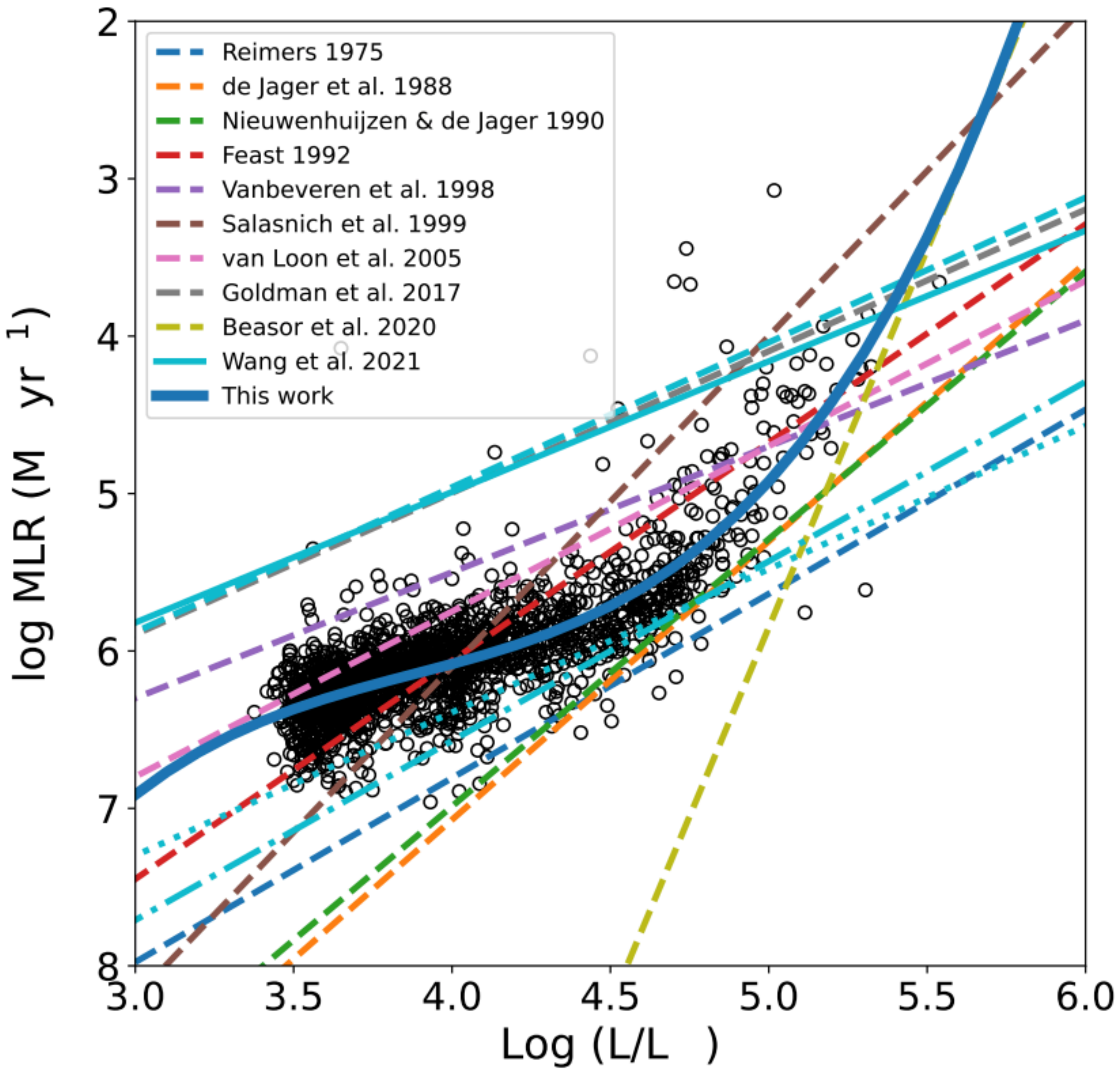


Massive stars nearing the late stages of their lives are called red supergiants because they are cool (red) and luminous because of their huge radii

Their winds are expected to be similar to AGB winds, but the wind strength is very uncertain!

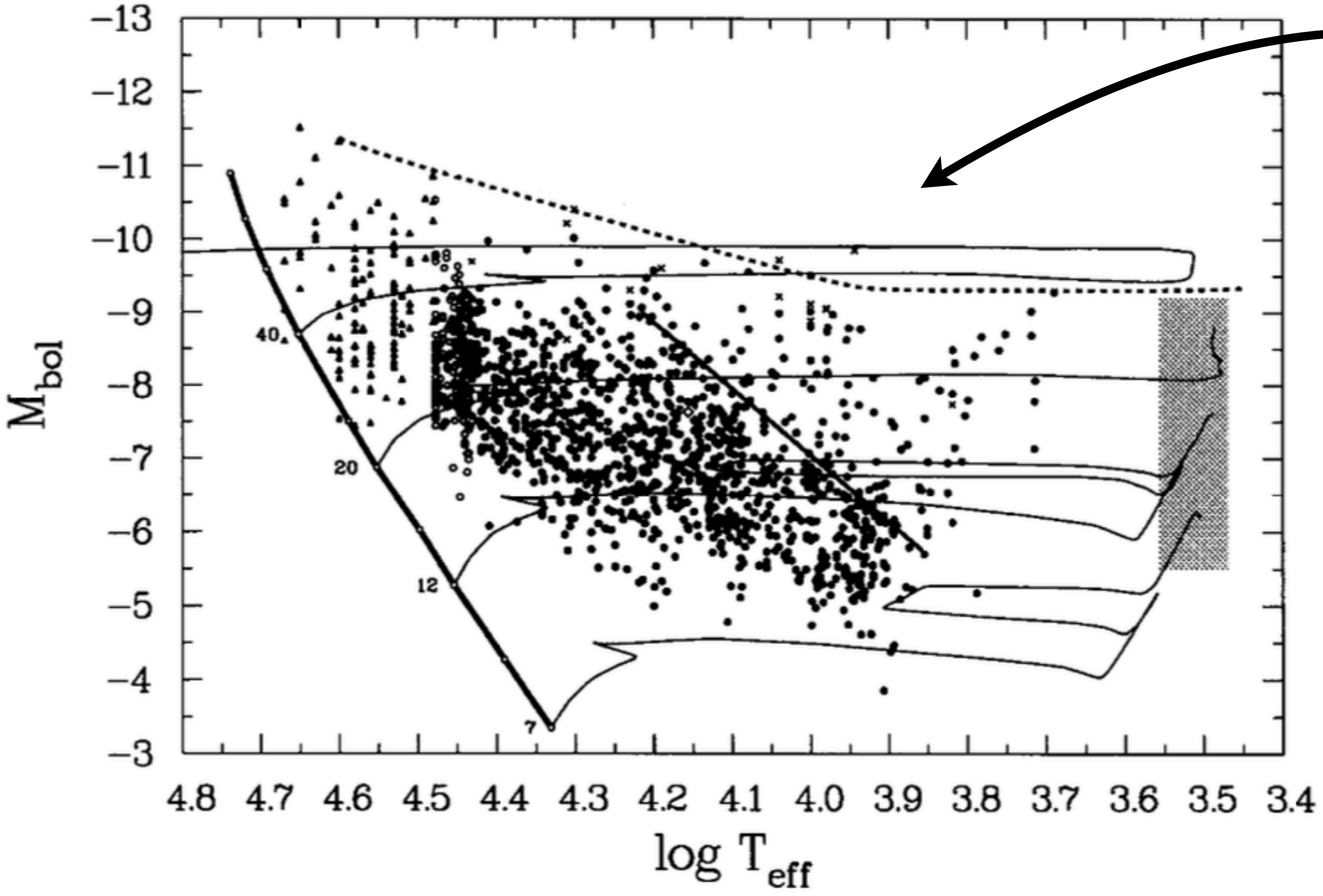


Beasor+2020,2023

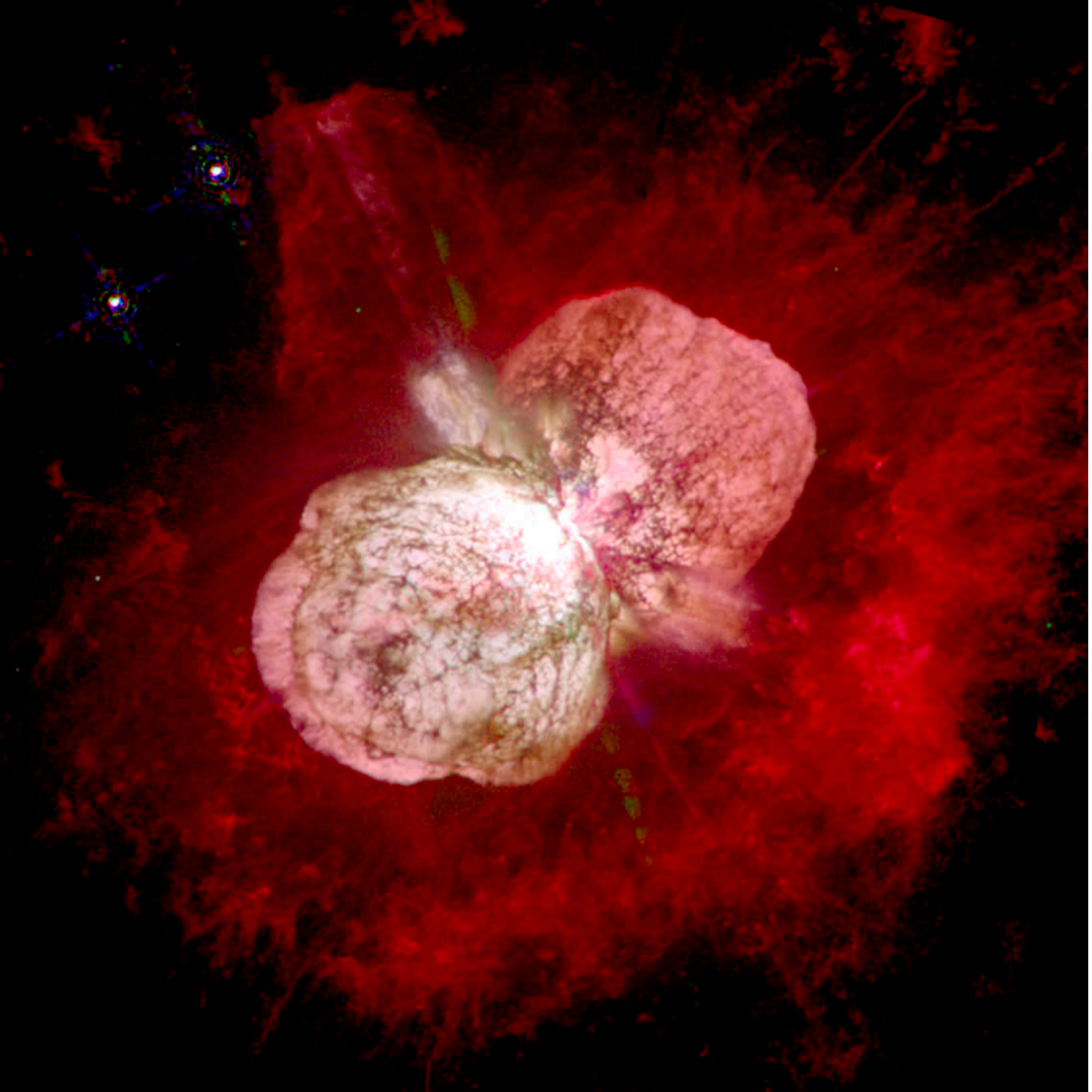


Yang+2023

Humphreys Davidson Limit



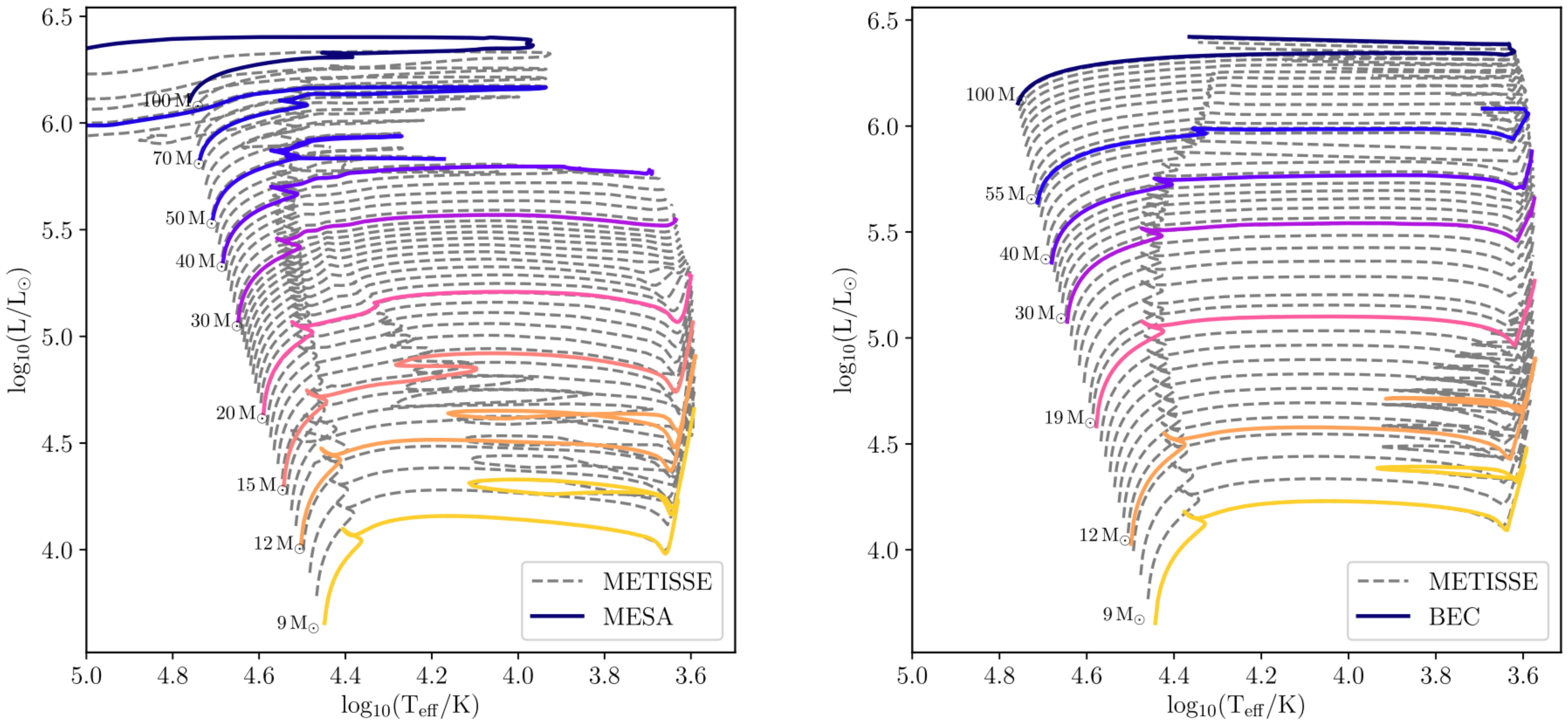
If stars existed for long periods of time here we would see them — but we don't!



Luminous blue variables

$$L_{\mathrm{Edd}} = \frac{4\pi c G M}{\kappa_{\mathrm{e}}},$$

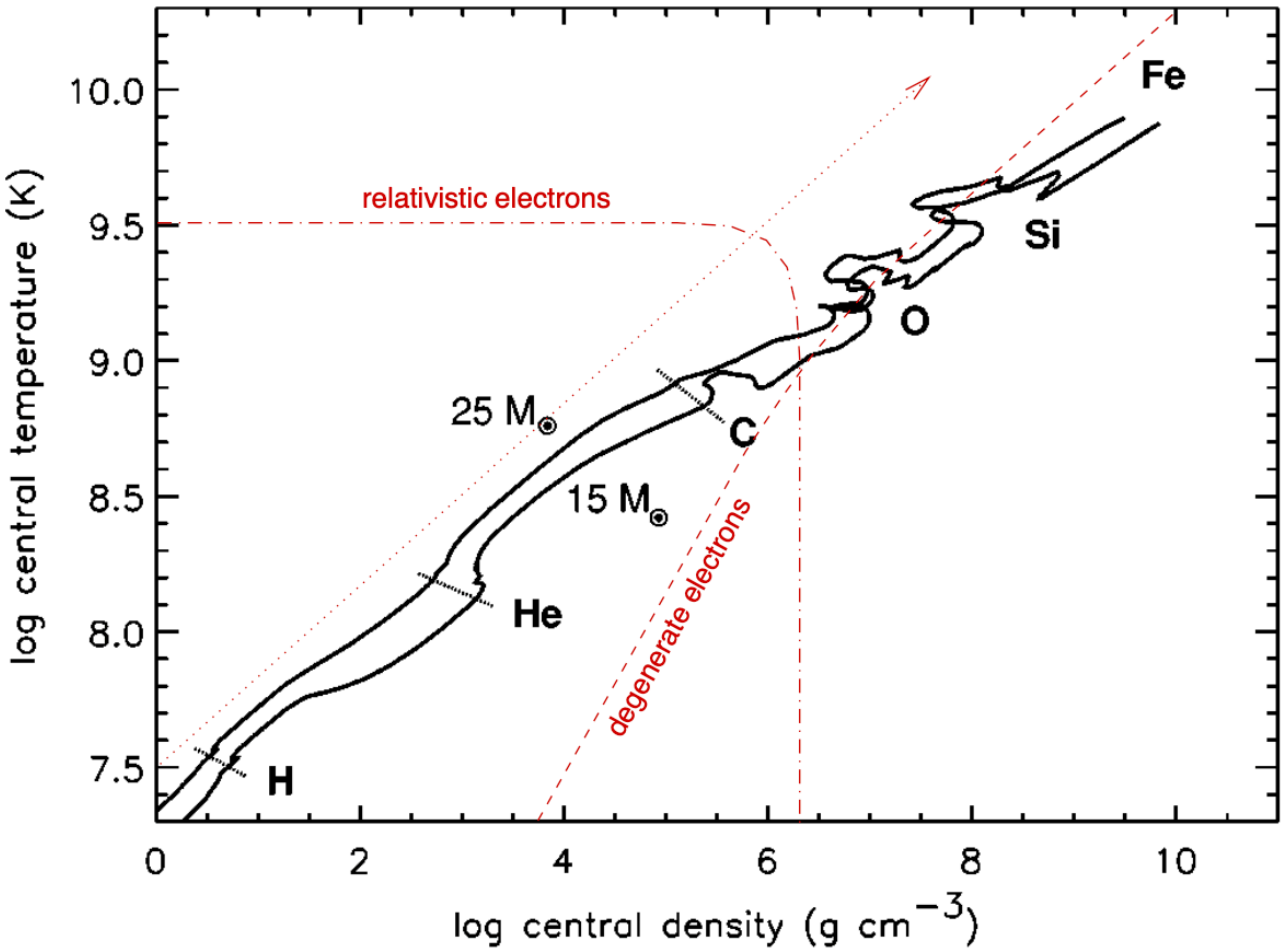
Huge mass eruptions in massive stars that can completely strip the envelope



Significant wind mass loss makes stars more blue (hotter)

—> Wolf Rayet objects (stripped envelope stars)

Late-stage nuclear burning

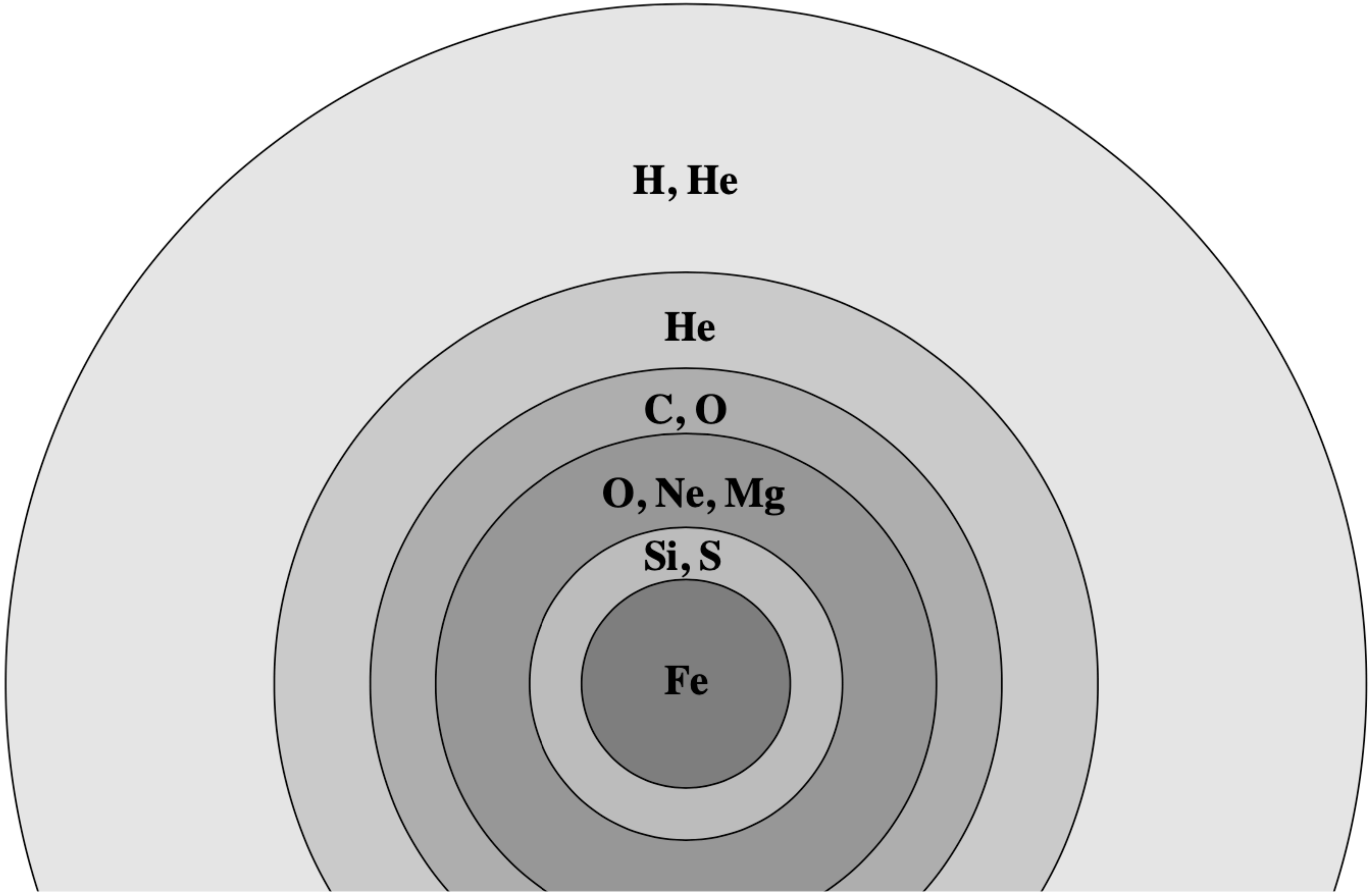


Evolution of core speeds up as temperature increases

This is because of accelerated neutrino emission

C burns for 1000 yrs
Si burns for ~1 day!

If stellar winds are not too strong, we end up with an onion-like structure and an Iron core



This final structure determines how the supernova explosion and/or compact object formation proceeds